KW 9: The Goods Market

Ch 3, Ch 5.1, Ch 14.1-14.2

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

CHAPTER 3: THE GOODS MARKET

Year-to-year movements in economic activity are driven by the interactions among production, income, and demand. Changes in demand lead to changes in production, which lead to changes in income, which in turn lead to changes in demand.

3-1 The Composition of GDP

GDP can be decomposed from the point of view of different buyers for goods. The terms 'output' and 'production' are synonymous.

The Five Components of GDP

Component	Symbol	US 2014 ($B)	% of GDP
Consumption	C	11,865	68.3%
Investment	I	2,782	16.0%
- Nonresidential		2,233	12.9%
- Residential		549	3.1%
Government spending	G	3,152	18.1%
Net exports	X-IM	-530	-3.1%
Inventory investment		77	0.4%
GDP (Y)		17,348	100.0%

Detailed Definitions

- **Consumption (C)**: Goods and services purchased by consumers. Largest component at 68% of GDP.
- **Investment (I)**: Also called fixed investment. Sum of nonresidential investment (firms buying plants/machines) and residential investment (people buying houses). NOT financial investment.
- **Government spending (G)**: Purchases of goods and services by government. Does NOT include government transfers (Medicare, Social Security) or interest on government debt. G = 18.1% of GDP, much less than total government spending (~33%) because transfers are excluded.
- **Net exports (X - IM)**: Exports minus imports. Trade surplus if X > IM; trade deficit if X < IM. U.S. had a trade deficit of 3.1% in 2014.
- **Inventory investment**: Difference between goods produced and goods sold. Production > Sales = positive inventory investment.

3-2 The Demand for Goods

Total demand for goods is denoted by Z:

$$Z \equiv C + I + G + X - IM$$

Three Simplifying Assumptions

- (1) All firms produce the same good (one market)
- (2) Firms supply any amount at a given price P (focus on demand; valid in short run)

- (3) Closed economy: no trade

$$X = IM = 0$$

Under closed economy assumption:

$$Z = C + I + G$$

Consumption (C)

- Consumption depends primarily on **disposable income**:

$$Y_D = Y - T$$

$$C = C(Y_D) \quad (+)$$

Linear specification:

$$C = c_0 + c_1 \cdot Y_D = c_0 + c_1(Y - T)$$

- **Marginal propensity to consume**: effect of an additional $1 of disposable income on consumption:

$$c_1 \quad \text{with restriction} \quad 0 < c_1 < 1$$

- Example: if c1 = 0.6, an additional $1 of income raises consumption by $0.60
- **Autonomous consumption**: consumption when disposable income is zero. Must be positive (people still need to eat):

$$c_0 > 0$$

- Changes in c0 reflect changes in consumer confidence, ease of borrowing, etc.

Investment (I)

- Treated as exogenous (taken as given) in this chapter:

$$I = \bar{I}$$

- This simplification is relaxed in Chapter 5 where investment depends on output and the interest rate

Government Spending (G)

- G and T together describe **fiscal policy**

- Both treated as exogenous: (1) governments don't behave with simple regularity, (2) we want to analyze policy scenarios

3-3 The Determination of Equilibrium Output

Combining all components:

$$Z = c_0 + c_1(Y - T) + \bar{I} + G$$

Equilibrium condition (production = demand):

$$Y = Z$$

Three Types of Equations in Models

1. Identities (e.g., disposable income definition)
2. Behavioral equations (e.g., consumption function)
3. Equilibrium conditions (e.g., production = demand)

Substituting demand into equilibrium:

$$Y = c_0 + c_1(Y - T) + \bar{I} + G$$

Using Algebra

Solving for Y:

$$Y = \frac{1}{1 - c_1}[c_0 + \bar{I} + G - c_1 T]$$

- **Autonomous spending**: the part of demand that does NOT depend on output

$$\text{Autonomous spending} = c_0 + \bar{I} + G - c_1 T$$

- **The Multiplier**: multiplies autonomous spending. Always > 1 because 0 < c1 < 1:

$$\text{Multiplier} = \frac{1}{1 - c_1}$$

- Example:

$$\text{if } c_1 = 0.6, \quad \text{multiplier} = \frac{1}{0.4} = 2.5$$

- A $1 billion increase in autonomous spending increases output by $2.5 billion

Using a Graph (Figure 3-2)

- Plot demand (ZZ line) and the 45-degree line (Y = Y)
- ZZ line: intercept = autonomous spending, slope = c1 (< 1, so flatter than 45-degree line)
- Equilibrium at point A where ZZ crosses the 45-degree line
- Left of A: demand > production. Right of A: production > demand.

The Multiplier Effect (Figure 3-3)

If c0 increases by $1 billion:

- Round 1: Demand increases by $1B -> Production rises $1B -> Income rises $1B
- Round 2: Consumption rises by c1 x $1B -> Production rises by c1 billion:

$$\Delta C_2 = c_1 \times \$1B$$

- Round 3: Consumption rises further:

$$\Delta C_3 = c_1^2 \times \$1B$$

- Total effect is a geometric series:

$$1 + c_1 + c_1^2 + \cdots + c_1^n \longrightarrow \frac{1}{1 - c_1} \quad (n \to \infty)$$

How Long Does Output Take to Adjust?

- In the model (no inventories, instant responses): adjustment is instantaneous
- In reality: adjustment takes time. Firms revise production quarterly, inventory adjustments occur first
- Output does not jump to Y' but increases gradually over time

Focus Box: Lehman Bankruptcy
During the 2008 crisis, consumption fell sharply despite disposable income initially not declining much. This represents a decrease in c0 (consumer confidence collapsed). Together with falling autonomous investment (housing), the multiplier amplified these into large output declines.

3-4 Investment Equals Saving: An Alternative Equilibrium View

Private Saving

$$S = Y - T - C = -c_0 + (1 - c_1)(Y - T)$$

- **Propensity to save**: how much of an additional dollar is saved:

$$(1 - c_1) = \text{propensity to save}$$

The IS Relation

Starting from the equilibrium condition, rearranging:

$$Y = C + I + G$$

$$I = S + (T - G)$$

- In equilibrium, **investment = private saving + public saving**
- This is called the **IS relation** (Investment = Saving)
- Public saving: T - G. If T > G: budget surplus. If T < G: budget deficit.

Paradox of Saving (Paradox of Thrift)

If consumers try to save more (c0 decreases):
- Equilibrium output DECREASES
- But total saving does NOT change (because I = I-bar is fixed)
- Mechanism: more saving -> less consumption -> less demand -> less output -> less income -> the fall in income offsets the attempt to save more
- Warning: only in the short run. In the medium/long run, higher saving can lead to higher investment and income.

3-5 Is the Government Omnipotent? A Warning

- Eq. (3.8) suggests the government can choose any output level. In reality:
- (1) Changing G or T is slow and contentious (legislative process)
- (2) Investment responds to output; imports leak demand abroad; exchange rates change
- (3) Expectations matter: consumer response depends on whether tax cuts are perceived as temporary or permanent
- (4) Side effects: too-high output can lead to increasing inflation
- (5) Budget deficits accumulate into public debt with long-run adverse effects

CHAPTER 5, SECTION 5-1: The Goods Market and the IS Relation

Investment Now Depends on Output and Interest Rate

In Chapter 3, investment was constant. Now investment depends on two factors:

- **(1) Level of sales/output (+)**: Higher sales -> firms need to invest to expand capacity
- **(2) Interest rate (-)**: Higher interest rate -> more costly to borrow -> less attractive to invest. Even with own funds, high i means it's more profitable to lend than to buy machines.

$$I = \underset{(+,\ -)}{I(Y,\ i)}$$

Determining Output

Replacing I-bar with I(Y, i) in the equilibrium condition:

$$Y = C(Y - T) + I(Y,\ i) + G$$

- Demand now depends on income Y (through C and I), interest rate i (through I), and fiscal policy (G, T)

Deriving the IS Curve

- For a given interest rate, the ZZ line shows demand as a function of output
- ZZ is upward sloping with slope < 1 (increase in demand < increase in output)
- Equilibrium where ZZ crosses 45-degree line

Figure 5-3: The IS Curve

- An increase in i decreases investment at any level of output -> ZZ shifts DOWN
- Equilibrium output DECREASES
- Plotting all (i, Y) pairs: the **IS curve is downward sloping**

IS Curve Properties

- Downward sloping: higher interest rate -> lower output
- Represents all (i, Y) combinations where goods market is in equilibrium
- Shifts RIGHT with: increase in G, decrease in T, increase in consumer confidence
- Shifts LEFT with: decrease in G, increase in T, decrease in confidence

CHAPTER 14: Financial Markets and Expectations

14-1 Expected Present Discounted Values

The expected present discounted value of a sequence of future payments is the value today of this expected sequence. It allows comparing a current cost against a stream of future benefits.

Basic Concepts

- $1 today is worth $(1 + i) next year (if lent at rate i)
- $1 next year is worth less than $1 today = the **present discounted value**:

$$\text{Present value of \$1 next year} = \frac{\$1}{1+i}$$

- i is the **discount rate**; the **discount factor** is:

$$\text{Discount factor} = \frac{1}{1+i}$$

- Higher interest rate -> lower present value of future payments
- Example: i = 5% -> $1 next year is worth $0.95 today
- Example: i = 10% -> $1 next year is worth $0.91 today

The General Formula

$$\$V_t = \$z_t + \frac{1}{1+i_t}\,\$z^e_{t+1} + \frac{1}{(1+i_t)(1+i^e_{t+1})}\,\$z^e_{t+2} + \cdots$$

- Each future payment is multiplied by its discount factor
- More distant payments have smaller discount factors -> lower present value
- Present value depends **positively** on future payments and **negatively** on interest rates

Special Cases

Constant Interest Rates

$$\$V_t = \$z_t + \frac{1}{1+i}\,\$z^e_{t+1} + \frac{1}{(1+i)^2}\,\$z^e_{t+2} + \cdots$$

- Weights decline geometrically:

$$1,\ \frac{1}{1+i},\ \frac{1}{(1+i)^2},\ \ldots$$

- i = 10%: payment in 10 years -> weight = 0.386; payment in 30 years -> weight = 0.057

Constant Rates and Payments (n years)

$$\$V_t = \$z \frac{1 - \left(\frac{1}{1+i}\right)^n}{1 - \frac{1}{1+i}}$$

- Example: 'Win $1M' lottery paying $50,000/year for 20 years at i=6%: actual present value is only ~$608,000

Perpetuity (Consol) - Payments Forever

$$\$V_t = \frac{\$z}{i}$$

- $10/year forever at i=5% -> worth $200. At i=10% -> worth $100

Zero Interest Rates

- If i = 0, present value = simple sum of all payments (useful approximation)

Nominal vs. Real Present Values

- **Nominal**: discount dollar payments using nominal interest rates
- **Real**: discount real payments (in terms of goods) using real interest rates
- The two approaches are equivalent:

$$\frac{\$V_t}{P_t} = V_t$$

- Use nominal for bond pricing; use real for consumption/investment decisions

14-2 Bond Prices and Bond Yields

Bond Basics

- Two key dimensions: **maturity** (time of payments) and **risk** (default risk, price risk)
- **Discount bonds** (zero-coupon): single payment at maturity (the face value)
- **Coupon bonds**: periodic coupon payments plus face value at maturity
- Coupon rate = coupon payment / face value
- Current yield = coupon payment / bond price
- **Yield to maturity**: the correct measure of return

Bond Market Vocabulary
- Government bonds (Treasuries, Bunds) vs Corporate bonds
- Bond ratings: Moody's Aaa to C; S&P AAA to D
- Risk premium: extra rate on risky bonds vs safest bonds
- Junk bonds: high default risk bonds
- T-bills: short-term government bonds (< 1 year)
- Consols: perpetual bonds (pay forever)
- TIPS: Treasury Inflation Protected Securities (indexed to inflation)

Bond Prices as Present Values

One-year discount bond (face value $100):

$$\$P_{1t} = \frac{\$100}{1 + i_{1t}}$$

Two-year discount bond (face value $100):

$$\$P_{2t} = \frac{\$100}{(1 + i_{1t})(1 + i^e_{1,t+1})}$$

Arbitrage and Bond Prices

- **Arbitrage**: if investors care only about expected returns (expectations hypothesis), expected returns must be equal across bonds

$$1 + i_{1t} = \frac{\$P^e_{1,t+1}}{\$P_{2t}}$$

- This implies: bond prices = expected present values of future payments

From Bond Prices to Bond Yields

- **Yield to maturity** for a 2-year bond: the constant annual rate making price = present value

$$\$P_{2t} = \frac{\$100}{(1 + i_{2t})^2}$$

Relationship between yield and short-term rates:

$$(1 + i_{2t})^2 = (1 + i_{1t})(1 + i^e_{1,t+1})$$

Approximation:

$$i_{2t} \approx \frac{1}{2}\left(i_{1t} + i^e_{1,t+1}\right)$$

- The 2-year rate is approximately the average of the current and expected future 1-year rate
- Generalizes: n-year rate = average of current and expected future 1-year rates over n years

The Yield Curve (Term Structure of Interest Rates)

- A graph of yield vs. maturity for bonds observed on a given day
- **Upward-sloping**: markets expect future short rates to RISE (e.g., expected economic expansion)
- **Downward-sloping (inverted)**: markets expect future short rates to FALL (e.g., expected slowdown)

Figure 14-2: U.S. Yield Curves

November 2000: Slightly downward sloping
- 3-month rate: 6.2%, 30-year rate: 5.8%
- Markets expected rates to decrease slightly (economy slowing)

June 2001: Steeply upward sloping
- 3-month rate: 3.5%, 30-year rate: 5.7%
- Fed had cut rates sharply; markets expected recovery and rate rises

SUMMARY OF KEY FORMULAS

(3.2) $C = c_0 + c_1 Y_D$ Linear consumption function

(3.3) $C = c_0 + c_1(Y - T)$ Consumption (income and taxes)

(3.8) $Y = \frac{1}{1 - c_1}[c_0 + \bar{I} + G - c_1 T]$ Equilibrium output

(3.10) $I = S + (T - G)$ IS relation

(3.11) $S = -c_0 + (1 - c_1)(Y - T)$ Private saving

(5.1) $I = I(Y, i)\ (+, -)$ Investment function

(5.2) $Y = C(Y - T) + I(Y, i) + G$ Goods mkt equilibrium

(14.1) $V = z + \frac{z^e}{1+i} + \cdots$ Present discounted value

(14.4) $\$P_{1t} = \frac{100}{1 + i_{1t}}$ 1-year bond price

(14.5) $\$P_{2t} = \frac{100}{(1 + i_{1t})(1 + i^e_{1,t+1})}$ 2-year bond price

(14.12) $i_{2t} \approx \frac{1}{2}(i_{1t} + i^e_{1,t+1})$ Yield approximation

$V = \frac{z}{i}$ Perpetuity value

$\frac{1}{1 - c_1}$ The Multiplier